An Overview of Three-Stem Audio Separation: An Analysis of the Cocktail Fork Problem

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I. Introduction

Modern multimedia materials like movies, podcasts, video games, and web videos often blend spoken words with background tunes and various noise elements into one complicated audio track. This blending makes it tough to manage or examine each part on its own. Researchers have long studied the classic cocktail party issue, which looks at how people zero in on a single sound amid all that noise. Still, most computer-based methods stick to simpler jobs, such as cleaning up voices or pulling apart instruments. They rarely tackle voices, tunes, and effects all at once. The study titled The Cocktail Fork Problem. Three-Stem Audio Separation for Real-World Soundtracks fills this void. It defines a way to split audio into three main parts, namely speech, music, and sound effects. The authors also present a top-notch collection of data called Divide and Remaster, or DnR for short. They pair it with a multi-resolution CrossNet model, known as MRX. This setup boosts how well the system handles actual audio tracks from media.

Putting this three-part splitting process into action gave a real sense of how choices in data preparation shape the model's results and its ability to work on everyday sound mixes. Things like picking sources, the way mixes happen, volume levels, and how sounds overlap all play key roles. When we tried to mimic parts of the DnR mixing method, it became clear that steady volume adjustments and true-to-life overlap patterns bring training examples much nearer to what you find in real shows or films.

Dealing with the MRX type of multi-resolution setup helped make sense of the balance between timing details and frequency details in audio work. It also explained why using different sizes for short-time Fourier transform windows helps grab quick bursts of effects alongside longer musical flows. Playing around with those window choices and various training options drove home some everyday issues in deep learning for audio tasks. These include how long to make data chunks, how to adjust learning rates over time, and limits on graphics processing unit memory.

Assessing the setup through measures like scale-invariant signal-to-distortion ratio across voice types and overlap cases built a stronger grasp of how numbers on a page connect to what the ear actually hears. It also tied into common needs, such as adjusting mixes or turning audio into text. In the end, this effort sharpened ideas around the cocktail fork challenge. It also developed hands-on know-how in setting up, teaching, and checking out current neural networks for pulling sounds apart.

II. Methodology

A. The Divide-and-Remaster Dataset

The Divide-and-Remaster dataset, often called DnR, gets put together by mixing three public collections of high-quality audio. These collections cover speech from LibriSpeech, music from the Free Music Archive, and sound effects from FSD50K. The standard LibriSpeech comes distributed at 16 kHz sampling rate. Still, the authors went ahead and rebuilt a version at 44.1 kHz. They did this by pulling from the matching high-fidelity recordings in LibriVox audiobooks. The choice of these three datasets really comes down to their compatible licensing terms, solid audio quality, and useful metadata that supports supervised learning tasks.

In order to produce mixtures that feel like real soundtracks, the authors set up a controlled pipeline for mixing. This pipeline draws on typical practices in audio production. Every mixture runs for a full 60 seconds. The setup ensures that clips from the same category avoid any temporal overlap. At the same time, overlaps between different categories arise in a natural way. The team adjusted loudness levels across classes through LUFS normalization. They followed standard conventions from film and broadcast audio in this approach. Speech tends to stand out louder than music or sound effects in those settings. They also took care to distinguish foreground from background sound effects. All of this helps the final mixtures maintain a sense of perceptual realism.

Through these methods, the dataset divides into 3406 mixtures for training, 487 for validation, and 973 for testing. Every mixture includes isolated ground-truth stems. These stems support training and evaluation of supervised models. [1].

B. Problem Formulation

Given a soundtrack mixture

$$y(t) = x_{\text{speech}}(t) + x_{\text{music}}(t) + x_{\text{SFX}}(t),$$

the aim of the separation model is to estimate the three source components. Most modern architectures operate in the short-time Fourier transform (STFT) domain, where the mixture is represented as a complex spectrogram Y. The model predicts real-valued masks $\hat{M}_j$ corresponding to each source class j,

and the time-domain estimate is obtained through masked inverse STFT:

$$\hat{x}_j = \mathrm{iSTFT}\Big(\hat{M}_j \odot Y\Big).$$

Model performance is evaluated using the scale-invariant signal-to-distortion ratio (SI-SDR), which provides a perceptually aligned measure of separation quality.

C. Multi-Resolution CrossNet (MRX)

The Multi-Resolution CrossNet (MRX) architecture extends the XUMX framework by incorporating multiple STFT resolutions. This design is motivated by the observation that speech, music, and sound effects display highly diverse temporal and spectral patterns. Short analysis windows provide superior time resolution, which is essential for capturing transient speech components, while long windows provide sharper frequency resolution that is advantageous for harmonic music structures and ambient SFX.

In MRX, the input mixture is transformed using several STFT window lengths. Each resulting spectrogram is projected into a common embedding dimension, after which the representations are averaged to produce a unified multi-resolution feature map. This combined representation is processed using bidirectional LSTM layers that model long-range temporal dependencies. The decoder then estimates a set of masks for each resolution and each source, and the final time-domain output is obtained by summing the inverse STFT reconstructions across all resolutions. This approach enables the model to jointly exploit fine-grain timing cues and wide-band frequency information.

D. Running the Pretrained Model

The provided codebase includes pretrained MRX weights and scripts for performing STFT analysis, mask inference, waveform reconstruction, and file handling. Using this implementation, we processed several mixtures to obtain separated speech, music, and SFX stems. The evaluation focused on listening quality, spectrogram inspection, behavior under heavy source overlap, and performance in silent or near-silent regions. These observations were compared with the documented results in the original paper to ensure consistency with expected model behavior.

III. Results and Discussion

A. Reported Results in the Paper

The published evaluation demonstrates that the Multi-Resolution CrossNet (MRX) model outperforms all baseline architectures, including Conv-TasNet, MaskTCN, UMX, and several single-resolution XUMX variants. When trained and tested at 44.1 kHz, MRX achieves SI-SDR gains of approximately 4.2 dB for music, 12.3 dB for speech, and 5.7 dB for sound effects. These improvements remain consistent across various overlap scenarios, underscoring the model's robustness in handling the complex interactions that occur in real-world soundtrack mixtures.

B. Evaluation Using a Custom Test Mixture

In addition to analyzing the results reported in the paper, I constructed an independent test mixture to evaluate the pretrained MRX model under conditions closely matching those used in the creation of the Divide-and-Remaster (DnR) dataset. The mixture was assembled in the Ableton digital audio workstation (DAW), following the guidelines described in the paper: each class (speech, music, sound effects) was selected from clean, isolated sources and normalized using LUFS-based loudness adjustments, with speech placed at a relatively higher level. Temporal structure was designed to mirror the DnR mixing procedure, including staggered clip placement, partial overlaps across classes, and non-overlapping behavior within each individual class. This simulated a realistic 60-second soundtrack-like environment comparable to those found in the DnR dataset.

The resulting mixture was exported at 44.1 kHz and passed through the pretrained MRX model to obtain the three estimated stems. The separation quality observed for this custom mixture aligned closely with the performance trends documented in the paper, providing additional validation that the model generalizes well to real, user-constructed content that follows similar mixing conventions.

C. Qualitative Analysis of Separated Sources

The speech stem produced by MRX exhibited strong intelligibility and minimal contamination from music or environmental sounds. Even during dense overlapping regions, the recovered speech remained perceptually clean, consistent with the high SI-SDR values reported for this class. For music, harmonic and melodic structures were preserved effectively, although light residual traces of speech harmonics occasionally appeared during segments with significant overlap. These artifacts were minor and expected given the spectral similarity between speech and musical content.

Sound effects exhibited the greatest variability, particularly for short transient events such as impacts or clicks. Ambient and continuous SFX were separated reliably, but sharp transients coinciding with strong musical onsets were sometimes smoothed—an outcome consistent with the heterogeneity of the SFX category and the difficulty highlighted in the paper when isolating highly unstructured content.

Silence regions were handled well by the model; residual energy was extremely low, reflecting the "Predicted Energy at Silence" (PES) behavior described in the original study. This demonstrates that MRX does not artificially introduce noise in non-active regions and maintains clean separation boundaries.

D. Benefits of Multi-Resolution Processing

The results from both the paper and the custom Ableton-generated mixture collectively reinforce the advantage of using multi-resolution analysis. Short STFT windows enabled the model to capture fine temporal dynamics such as speech consonants and rapid SFX transients, while longer windows

allowed accurate reconstruction of stable harmonic components present in music and background ambience. By integrating information across multiple resolutions, MRX produced smoother, more coherent masks and reduced artifacts that typically arise from single-resolution models. This multi-resolution strategy ultimately contributed to balanced and reliable performance across all three source types, further validating its relevance for real-world soundtrack separation.

IV. Conclusion

This report examined the Cocktail Fork Problem and reproduced the pretrained MRX model's outputs. The paper makes a strong contribution through both the dataset design and a principled multi-resolution architecture. Experimental analysis confirms that MRX reliably separates real-world soundtrack mixtures at full bandwidth.

Potential future directions include integrating source separation with downstream ASR/SFX classification, using perceptual loss functions, and extending to multi-channel spatial audio.

References

[1] D. Petermann, G. Wichern, Z.-Q. Wang, and J. Le Roux, "The Cocktail Fork Problem: Three-Stem Audio Separation for Real-World Soundtracks," in *Proc. ICASSP*, 2022. (Information sourced from the provided PDF.)